Static Data Members:

When the data member of a class declared as static , it is commonly shared by all the objects created for that class. Static data member is also called as class variable. Static data member is initialized to zero when the first object is created, if no other initialization is present.

Any changes in the static data member through one member function will reflect in all other object's member functions.

Declaration:

```
static data_type member_name;
```

Defining the static data member

It should be defined outside of the class following this syntax:

```
data_type class_name :: member_name =value;
```

Example :

```
#include<iostream>
using namespace std;
class item
{
static int count;   // static data member declaration
public:
void DisplayCounter()
{
count++;
cout<<"count:"<<count<<endl;
}
};
int item::count;   // static data member definition
int main()
{
item a,b,c;
a.DisplayCounter();
b.DisplayCounter();
c.DisplayCounter();
return 0;
}
Output:
Count: 1
```

Vector India

```
Count: 2
Count: 3
```

Static Member Function:

Like data member of the class, member function of the class can also be declared as a static.
Static member functions are designed to work with static data members.
In order to make a member function as static we need to precede the function declaration with static keyword.
Static member function having several characteristics which are given below:
(1) Static member function can access only static member of the class in which it is declared.
(2) Static member function is not a part of class object so it can not be called using object of the class.

Static member function can be called using class name and scope resolution operator as shown below:

```
Class_Name :: Function_Name ();
```

Example :

```
#include <iostream.h>
#include <conio.h>
class test
{
private:
static int count;
public:
void setCount(void);
static void DisplayCounter(void);
};
void test :: setCount(void)
{
count++;
}
void test :: DisplayCounter(void)
{
cout<<"Count:"<< count << endl;
```

Vector India

```
}
int test::count;
int main(void)
{
test t1, t2;
clrscr();
t1.setCount();
test :: DisplayCounter();
t2.setCount();
test :: DisplayCounter();
getch();
return(0);
}
Output:
Count: 1
Count: 2
```

Vector India